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Research on tissue section negative detection algorithm based on multispectral microscopic imaging

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In recent years, the rapid advancement of artificial intelligence (AI) technology has enabled AI-assisted negative screening to significantly enhance physicians' efficiency through image feature analysis and multimodal data modeling, allowing them to focus more on diagnosing positive cases. Meanwhile, multispectral imaging (MSI) integrates spectral and spatial resolution to capture subtle tissue features invisible to the human eye, providing high-resolution data support for pathological analysis. Combining AI technology with MSI and employing quantitative methods to analyze multiband biomarkers (such as absorbance differences in keratin pearls) can effectively improve diagnostic specificity and reduce subjective errors in manual slide interpretation. To address the challenge of identifying negative tissue sections, we developed a discrimination algorithm powered by MSI. We demonstrated its efficacy using cutaneous squamous cell carcinoma (cSCC) as a representative case study. The algorithm achieved 100% accuracy in excluding negative cases and effectively mitigated the false-positive problem caused by cSCC heterogeneity. We constructed a multispectral image (MSI) dataset acquired at 520 nm, 600 nm, and 630 nm wavelengths. Subsequently, we employed an optimized MobileViT model for tissue

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classification and performed comparative analyses against other models. The experimental results showed that our optimized MobileViT model achieved superior performance in identifying negative tissue sections, with a perfect accuracy rate of 100%. Thus, our results confirm the feasibility of integrating MSI with AI to exclude negative cases with perfect accuracy, offering a novel solution to alleviate the workload of pathologists.

Keywords: Multispectral imaging; artificial intelligence; cSCC; negative detection.

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1. Introduction

Cancer remains the second leading cause of death worldwide, with incidence rates showing a persistent upward trend. Globally, there were approximately 20 million new cases and 10 million deaths in 2022, with projections indicating a dramatic increase to 35 million cases by 2050 — representing a 77% growth compared to 2022.[1] Histopathology serves as the "gold standard" for disease diagnosis, yet pathologist resources supporting this standard face severe global shortages and overwhelming workloads. China's 2022 data revealed only 20,400 registered pathologists, each handling an exceptionally high workload of approximately 7000 diagnostic cases annually,[2] which clearly demonstrates the tremendous pressure on these medical professionals. The substantial workload burden stems primarily from the extremely high proportion of negative samples in cancer screening. A large-scale Chinese study of 15,996 individuals undergoing health examinations found a lung cancer detection rate of merely 2.1%,[3] while Australian 2023 data showed an equally low cancer detection rate of 3.8% in screening programs.[4] Consequently, pathologists must devote significant time and effort toward examining negative screening samples, an approach that also suffers from limitations such as inter-observer variability[5] and visual fatigue.[6] Therefore, developing efficient and reliable "negative exclusion technology" (automated identification and exclusion of definitively negative samples) could liberate pathologists from the burdensome task of negative sample screening, allowing them to focus on in-depth analysis of high-risk and complex cases. This methodology has the potential to substantially enhance both the efficiency and quality of diagnostics. Empirical evidence indicates that robust negative exclusion techniques can reduce false-negative rates by 27–35%.[7–9] The evolution of automated negative exclusion technology has progressed through several distinct phases, beginning with artificial intelligence (AI) analysis reliant on RGB imaging. Early computer-aided diagnostic systems primarily utilized conventional RGB images; however, this approach is inadequate for capturing deep spectral features such as hemoglobin absorption profiles and molecular specificity, as it results in limited sensitivity and elevated false-negative rates.[10,11] A fundamental constraint lies in the inherent spectral overlap within RGB channels, which restricts precise biochemical quantification.

Subsequent advancements have been ushered in by multispectral imaging (MSI), which integrates spectral and spatial data to overcome the resolution limitations of traditional histopathology. The core strength of MSI is its capacity to resolve tissue heterogeneity at the subcellular level by capturing wavelength-specific spectral responses. This capability enables the quantitative analysis of molecular constituents — for instance, discriminating nuclear from cytoplasmic components,[12] classifying cancer subtypes like pulmonary adenocarcinoma versus squamous cell carcinoma (cSCC),[13] and differentiating tissue types such as collagenous fibers in hepatic sections.[14] Li *et al.*[15] comprehensively summarized the key advances in both *in vivo* and *ex vivo* applications of this technology. Building upon this foundation, our group's preliminary work has further demonstrated the potential of MSI in grading cSCC by evaluating morphological features including nuclear-to-cytoplasmic ratio and keratin pearl formation[16] — a finding that directly motivates the current study. When coupled with spectral unmixing algorithms, MSI also facilitates the simultaneous detection of multiple biomarkers.[17] The continued integration of microscopy and spectroscopy, enabling continuous spectral acquisition at single-cell resolution, has expanded its applications to both *in vivo* contexts (e.g., assessing cutaneous burn depth[18] and screening for oral potentially malignant disorders[19])

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and *ex vivo* analyses (e.g., distinguishing histological staining methods[20] and identifying early neoplastic lesions[21]).

The most recent phase in this technological trajectory is characterized by the integration of MSI with AI, particularly deep learning. This confluence has established a novel paradigm wherein AI models autonomously extract spectrospatial features to directly correlate pathological characteristics with diagnostic outcomes. This synergy is evidenced by the rapid translation of spectral technologies from laboratory validation to clinical decision support, with intelligent whole-slide image analysis frameworks now being deployed in areas such as tumor immunotherapy response prediction and cervical cancer subtyping. [22–24] These systems have yielded remarkable progress, including reduced analytical time for cervical cancer slides[25] and lower rates of missed diagnosis in breast cancer screening.[26] Furthermore, automated pipelines for stain normalization and lesion annotation have been developed, thereby facilitating the construction of large-scale, curated datasets.[27]

Despite these advancements, extant research — including our preliminary findings — indicates that the collaborative efficacy of AI and MSI remains insufficient to achieve the ideal clinical benchmark of 100% negative predictive value. For instance, the AstroPath platform, while achieving a high AUC (> 0.85) in predicting immunotherapy response, demonstrates a negative exclusion efficiency of 95%, leaving a 5% risk of missed detections.[28] Similarly, an SVM-based spectral analysis of bladder tumor cells attained 96% sensitivity but exhibited a 10% misclassification rate in negative samples.[29] The optimized SOD-GAN framework reached 97% accuracy in cervical cancer classification, yet its performance declines in extended staging tasks.[30] While AI can exclude 70% of negative slides, in cytological prescreening, a substantial 30% manual review burden persists, alongside suboptimal sensitivity for detecting low-abundance lesion cells.[23]

To holistically improve negative exclusion rates while maintaining high diagnostic specificity, these systems must advance in analyzing complex heterogeneous tissues, enhancing ultra-weak spectral signals, and improving feature generalization from limited samples. This study presents an efficient negative pathology screening and grading method for cSCC tissue sections by integrating a custom-designed automated multispectral microscopic imaging system with deep learning and AI algorithms.

2. Materials and Methods

2.1. *Experimental samples*

This study was conducted using a cohort of 1083 independently sourced H&E-stained tissue sections of cutaneous squamous cell carcinoma (cSCC) and corresponding normal skin procured from Ruijin Hospital, Shanghai Jiao Tong University. The dataset comprised 440 normal tissue sections, 343 highly differentiated cSCC sections, and 300 poorly differentiated cSCC sections, with all specimens derived from distinct pathological cases to maximize data diversity and statistical independence. Each tissue section was rigorously evaluated, diagnosed, and staged by two senior pathologists to establish a definitive histopathological ground truth. The study protocol was approved by the Ethics Committee of Ruijin Hospital, Shanghai Jiao Tong University, and written informed consent was obtained from all participants or their legal guardians.

2.2. *Multispectral image acquisition and correction*

2.2.1. *Experimental equipment*

In our preliminary study, we developed an automated multispectral microscopic imaging system featuring efficient automatic image acquisition, cost-effective implementation, and a simplified structural design. The system consisted of three core modules: (1) Illumination Module: The light source system comprised 13 narrow-band LED illumination units covering a spectral range of 420–680 nm with a 20 nm bandwidth, ensuring comprehensive coverage of the visible spectrum. The switching speed between the LED channels was electronically controlled to be less than 1 ms, which was negligible compared to the camera exposure time, thereby eliminating potential temporal artifacts during spectral sequencing. (2) Microscopic Imaging Module: This module was composed of a CCD camera (Pike F-145B, Allied Vision Technologies, Germany) and two microscope objectives (numerical aperture (NA) = 0.35, magnification × 20; NA = 0.14, magnification × 5; PDV, China) to

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meet the fundamental requirements for imaging clarity and magnification. The field of view (FOV) is determined to be 1.79 mm × 1.34 mm and 0.45 mm × 0.34 mm for 5× and 20× objectives, respectively, calculated based on the camera sensor size and objective magnification. (3) Motorized Translation Module: A two-dimensional (2D) motorized stage system (XY-5050, Zolix, China), integrated with high-precision stepper motors and drivers, enables automated sample scanning and precise positioning. The Qt-based software interface facilitated automatic multispectral image acquisition by controlling stage movement with predefined step sizes. To ensure imaging accuracy, the software introduced a settling delay after LED switching and before camera triggering to guarantee a stable illumination. This integrated system simplified operation and supported multiband imaging and rapid large-area scanning, making it particularly suitable for large-area tissue slice applications that require high-quality simultaneous spatial and spectral data acquisition. This modular architecture ensured excellent system scalability for future enhancements, as detailed in our previous work,[31] with the system schematic shown in Fig. 1, hardware specifications in Table 1, and performance metrics in Table 2.

The image acquisition procedure was as follows: First, the system sequentially activated 13 narrowband LED sources (420–680 nm, 20 nm intervals), incorporating a brief stabilization delay after each switch. For each wavelength, the high-resolution CCD camera captured a monochromatic image. Concurrently, the motorized translation stage repositioned the sample according to a predefined path to ensure complete spatial coverage. This cycle was repeated until a full spectral image cube of the entire slide was obtained. Finally, all captured images underwent dark-current and flat-field correction to mitigate noise and correct for nonuniform illumination, thereby guaranteeing the fidelity of the dataset. Based on this procedure, the estimated acquisition time for a multispectral image stack (13 bands) in a single FOV was approximately 3–5 s, demonstrating the system's capability for rapid data capture.

2.2.2. *Multispectral image acquisition and correction*

To ensure imaging accuracy and consistency, all acquisition procedures were conducted in a controlled darkroom environment under the supervision of two experienced pathologists. For each lesion area, we acquired 13 grayscale images at distinct spectral bands; sequential scanning of different slide regions was then performed by controlling the mechanical stage translation. Exposure time was calibrated for each wavelength to ensure consistent acquisition parameters. Imaging was performed on all 1083 tissue sections; each region was imaged five times, and grayscale values were averaged across replicates, thereby reducing single-acquisition variability. All the spectral images collected underwent dark current and flat-field correction to ensure consistency of illumination for different samples. The correction method is as follows:

$$R(\lambda) = \frac{I_{raw}(\lambda) - I_{\text{dark}}}{I_{\text{ref}}(\lambda) - I_{\text{dark}}}. \tag{1}$$

Here, $I_{\text{raw}}(\lambda)$ denotes the sample image intensity, I_{dark} represents the background intensity under no

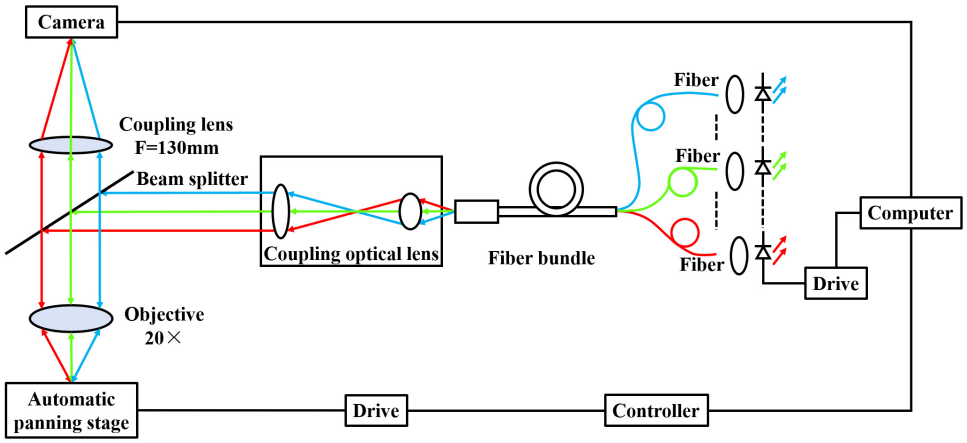


Fig. 1. The overall working principle of image acquisition equipment.

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Table 1. Hardware specifications parameters.

Component	Parameter	Specification
LED light source	Type	High-power LED
	Power	3 W
	Wavelength range	420–680 nm
	Operating voltage	2.0–2.2 V
	Drive current	700 mA
	Luminous power	300–400 mW
	Beam angle	120°
	Package size	8 × 5.5 mm
CCD camera	Sensor type	Sony ICX285 CCD (progressive scan)
	Resolution	1388 × 1038 pixels
	Pixel size	6.45 μm × 6.45 μm
	FOV	With 5× objective: 1.79 mm × 1.34 mm
		With 20× objective: 0.45 mm × 0.34 mm
	Frame rate	30 fps
	Spectral response	400–1000 nm
	Dynamic range	73 dB
Motorized translation stage	Model	Zolix XY-5050 2D Motorized Stage
	Travel range	50 mm × 50 mm
	Positioning accuracy	± 1 μm
	Minimum Step Size	0.156 μm (256 microstepping)
	Maximum Velocity	30 mm/s

illumination, and $I_{\text{ref}}(\lambda)$ stands for the reference image intensity under different wavelength illuminations.

2.3. *Image preprocessing*

To ensure a rigorous and unbiased evaluation of model performance, the dataset was partitioned at the patient level into training (70%), validation (20%), and test (10%) subsets. To ensure a robust performance of the deep learning models — particularly given the parametric demands of the Transformer-based architecture — and to mitigate the overfitting risks associated with the available sample size, an extensive data augmentation strategy was implemented [Fig. 2]. The original 1388 × 1038-pixel images were first center-cropped to 1024 × 1024 pixels in order to retain essential pathological structures while conforming to model input requirements. The training set was subsequently augmented in real-time through a series of geometric transformations, including rotation at 90° increments, horizontal and vertical flipping, and scaling at factors of 0.8× and 1.2×. As summarized in Table 3, this augmentation strategy expanded the total number of training instances by a factor of eight, effectively enriching feature variability and supporting stable model convergence.

Table 2. System performance parameters.

Number of bands	Wavelength (nm)	Spectral resolution (nm)	Spatial resolution (μm)	Translational table stroke (mm)
13	420–680	20	≤ 1.1	50 × 50

2.4. *Classification model*

2.4.1. *MobileViT*

The MobileViT[32] model, which is a lightweight vision Transformer for mobile devices, was introduced by Google in 2021. It combines the advantages of CNN and ViT to address traditional CNN's global information processing limitations and ViT's high computational complexity on resource-constrained devices.[33] As a lightweight vision Transformer, MobileViT enables efficient mobile inference through collaborative CNN and Transformer design.

Figure 3 illustrates the overall network architecture of MobileViT. By progressively downsampling feature maps (e.g., 512 × 512 → 128 × 128) to reduce computational cost, the architecture integrates 3 × 3 convolutions and low-rank projections to limit parameter growth while utilizing implicit 2 × 2 patch partitioning to optimize memory usage.

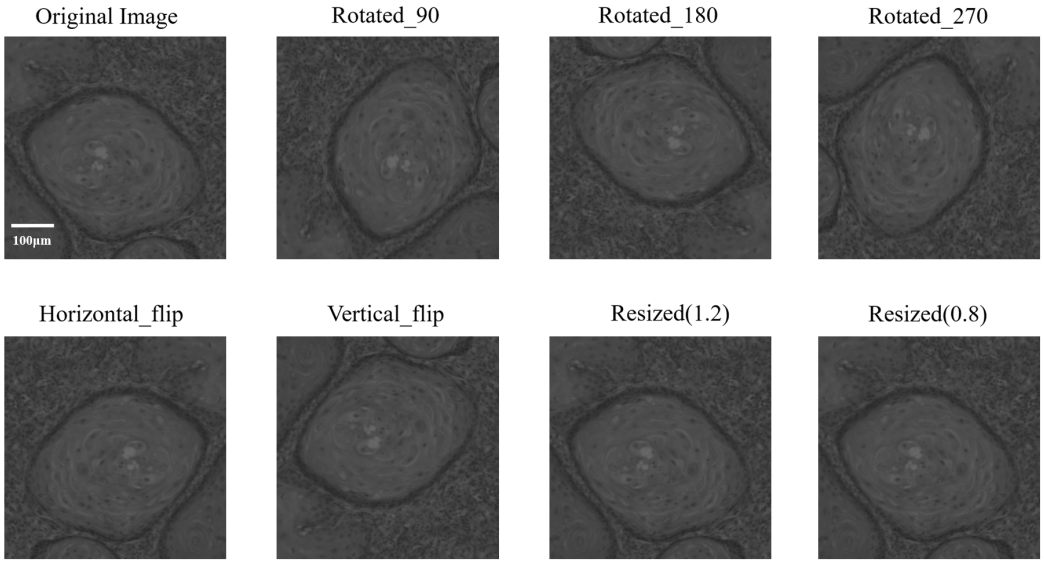


Fig. 2. Multispectral image expansion.

Table 3. Multispectral dataset partitioning.

	Train	Val	Test	Expand
Low	211	29	60	1688
High	240	34	69	1920
Normal	309	44	87	2472

Finally, the network reduces dimensions through global average pooling and produces classification via a single, fully connected layer. The downsampling strategy and compact module design are inherently suited to resource-constrained environments, achieving memory efficiency without explicit patch processing.

Figure 4 highlights the core MobileViT Block. It effectively combines the strengths of CNN and ViT through three phases: local feature modeling, global feature modeling, and feature fusion. In the local feature phase, a 3×3 convolution captures low-level patterns such as edges and textures. The processing begins with feature map X (size (C, H, W)), applies a 3×3 convolution for local feature extraction, and outputs feature maps of size (d, H, W), as expressed

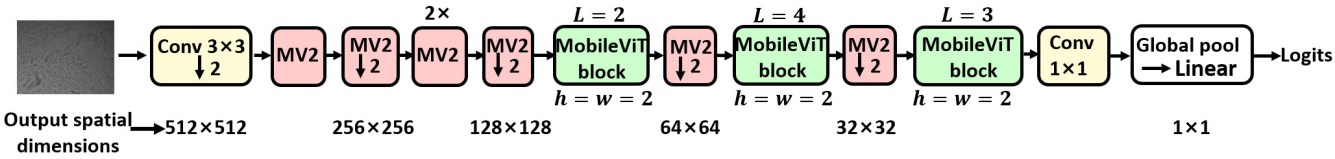


Fig. 3. MobileViT model structure diagram.

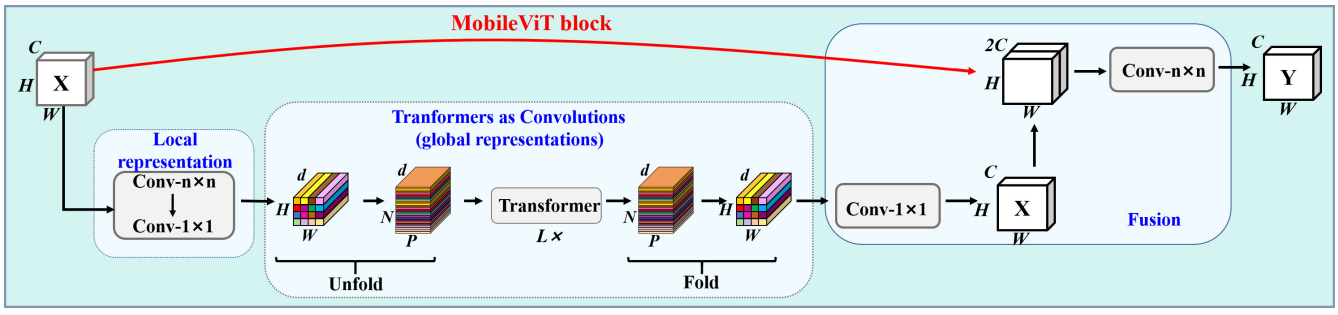


Fig. 4. The overall block diagram of MobileViT block.

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below:

$$F_{\text{local}} = \sigma(W_{\text{conv}} * X + b). \quad (2)$$

Here, W_{conv} is the convolutional kernel weight matrix, b is the bias term, and σ is the activation function.

Global feature extraction leverages the Transformer's global modeling capability using a patching strategy to reduce computational complexity. The specific algorithm is as follows:

$$Z' = \text{Unfold}^{-1}(\text{Transformer}(\text{Unfold}(F_{\text{local}}))). \quad (3)$$

The input is the local feature map F_{local}(dimension $H \times W \times d$) extracted by the convolutional layer, and the output is the feature map Z' (also dimension $H \times W \times d$) integrated with global information. The core objective of this process is to enable each spatial location (or small region) in the feature map F_{local} to comprehend the entire image context. The Unfold operation divides the feature map into a sequence of nonoverlapping local patches. Each patch is independently fed into the Transformer module. Inside the Transformer, the self-attention mechanism establishes global semantic connections between patches, modeling cross-regional visual dependencies. Finally, the Fold operation restores the processed sequence to a spatial feature map structure, maintaining the original image topology.

During the feature fusion stage, the original input features and the globally processed features are concatenated along the channel dimension. The convolutional layer then integrates local details with global context information.

Through this design, the MobileViT Block fully utilizes fine-grained and global context information in images while maintaining computational efficiency. This enhances the performance of the MobileViT model across various visual tasks.

2.4.2. *Loss function*

To address the significant class imbalance in our dataset (Normal: 40.66%, Highly-differentiated: 31.58%, Poorly-differentiated: 27.77%), we replaced the standard cross-entropy loss[34] with adaptive focal loss.[35] While cross-entropy loss is widely used in classification tasks, we observed during preliminary experiments that it biased the model toward the majority class (normal tissue), resulting in suboptimal recognition of tumor subtypes. To mitigate this issue, we employed focal loss, which introduces a modulating factor that reduces the relative loss for well-classified examples, thereby focusing training on challenging minority classes. The loss function is formulated as

$$FL(p_t) = -\alpha_t(1 - p_t)^{\gamma} \log(p_t). \quad (4)$$

Here, p_t represents the predicted probability of the current class and $(1 - p_t)^{\gamma}$ attenuates the loss for high-confidence samples by focusing more on low-confidence, hard-to-classify samples during training. This helps enhance the model's ability to recognize minority classes, especially under conditions of class imbalance.

Additionally, we introduce a learnable weight coefficient, α, which dynamically adjusts in inverse proportion to class frequency.

$$\alpha_i = \frac{1}{\text{category ratio}_i} \Bigg/ \sum_{j=1}^{3} \frac{1}{\text{category ratio}_j}. \quad (5)$$

The weight coefficient α assigns greater weights to minority classes (such as highly differentiated and poorly differentiated cSCC), causing the model to pay more attention to these class samples during training. This reduces the model bias caused by class imbalance and improves the overall classification performance.

This two-stage loss function optimization strategy enhances the focus on hard-to-classify samples and improves the model's ability to recognize minority classes.

2.4.3. *Activation function*

To enhance the model's capacity to learn fine-grained features from our imbalanced multispectral data, we replaced the standard rectified linear unit (ReLU) activation with the Mish function. The MobileViT model typically employs the ReLU activation function. However, the gradient of ReLU is zero in the negative domain, potentially leading to dead neurons, which results in diminished contributions from features in the negative range to the model. The Mish activation function, proposed by Misra,[36] is mathematically expressed as follows:

$$\text{Mish}(x) = x \cdot \tanh(\ln(1 + e^x)) \quad (6)$$

As shown in Eq. (6), the Mish activation function maintains a nonzero gradient across its entire domain. Moreover, the self-regulating property (*tanh* component) of Mish adaptively regulates the flow of

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negative information, preserving more subtle features such as cellular nucleus morphological differences, thereby enhancing the model's ability to capture fine textures indicative of cSCC differentiation levels. Additionally, Mish has a larger dynamic range, allowing for a more thorough transmission of input information between layers and enhancing the model's adaptability to complex data distributions. This characteristic improves the model's generalization capability, making it more accurate and reliable in tasks such as pathological image classification. Therefore, this experiment chooses the Mish activation function to optimize the model.

2.5. *Evaluation method*

This study evaluates the negative prediction model using accuracy (Acc), precision (Pre), recall (Re), specificity (Spe), and $F1$-score. The specific calculation methods are as follows:

$$\mathrm{Acc} = \frac{\mathrm{TP} + \mathrm{TN}}{\mathrm{TP} + \mathrm{TN} + \mathrm{FP} + \mathrm{FN}} \times 100\%, \tag{7}$$

$$\mathrm{Pre} = \frac{\mathrm{TP}}{\mathrm{FP} + \mathrm{TP}} \times 100\%, \tag{8}$$

$$\mathrm{Re} = \frac{\mathrm{TP}}{\mathrm{FN} + \mathrm{TP}} \times 100\%, \tag{9}$$

$$\mathrm{Spe} = \frac{\mathrm{TN}}{\mathrm{TN} + \mathrm{FP}} \times 100\%, \tag{10}$$

$$F1 = 2 \times \frac{\mathrm{Precision} \times \mathrm{Recall}}{\mathrm{Precision} + \mathrm{Recall}} \times 100\%. \tag{11}$$

Here, TP stands for True Positives, indicating the number of samples correctly predicted as positive by the model (such as tumor tissues). TN represents True Negatives, indicating the number of samples correctly predicted as negative by the model (such as normal tissues). FP denotes False Positives, indicating the number of samples incorrectly predicted as positive by the model (actually normal tissues). FN represents False Negatives, indicating the number of samples incorrectly predicted as negative by the model (actually tumor tissues). The accuracy ranges from 0 to 1, with higher values indicating a better overall classification performance of the model. An accuracy of 1 implies that the model predicts all samples correctly, whereas an accuracy of 0 indicates that the model's predictions are entirely incorrect.

3. Result

3.1. *Multispectral data analysis*

In this study, we observed that MSI offers distinct advantages over conventional RGB imaging in the analysis of pathological images of cSCC. The pathological sections in this study were stained with hematoxylin and eosin (H&E). Hematoxylin stains cell nuclei dark blue or black, with a primary absorption peak around 560 nm and significant absorption across the 400–600 nm range, whereas eosin stains cytoplasm and connective tissue pink or red, exhibiting primary absorption peaks at 518 nm and 528 nm along with strong absorption characteristics in the 450–650 nm range. Based on the optical properties of H&E staining, this study selected three key wavelengths — 520 nm, 600 nm, and 630 nm — for MSI. As shown in Fig. 5, imaging at these three specific wavelengths highlights distinct histological features: the 520 nm band near the primary absorption peak of eosin enhances the contrast of collagen fibers and eosin-stained lesional epidermis, making them more distinct and facilitating differentiation between normal and tumor tissues; the 600 nm band leverages the weak absorption of eosin to enhance the contrast of large lipid droplets in normal tissues as well as keratin pearls and squamous eddies through refractive index differences while suppressing nuclear signals to highlight extracellular matrix features; the 630 nm band, located on the long-wavelength side of hematoxylin's absorption peak where eosin absorption has diminished, provides optimal contrast under these absorption characteristics, emphasizing nuclear features in the images. In contrast, RGB imaging suffers from overlapping optical signals, as the absorption peaks of both H&E fall within the green channel (500–600 nm) of RGB, making it impossible to distinguish whether the signals originate from biomolecules or staining agents. Therefore, selecting these three specific wavelengths for imaging allows for the full utilization of the optical properties of H&E staining, significantly improving the identification of pathological tissue structures.

By focusing on 520, 600, and 630 nm spectral bands, this study leverages the unique optical properties of H&E staining to enhance tissue classification in cSCC pathology. We systematically acquired a comprehensive multispectral image dataset encompassing poorly differentiated cSCC, highly differentiated cSCC, and normal tissue

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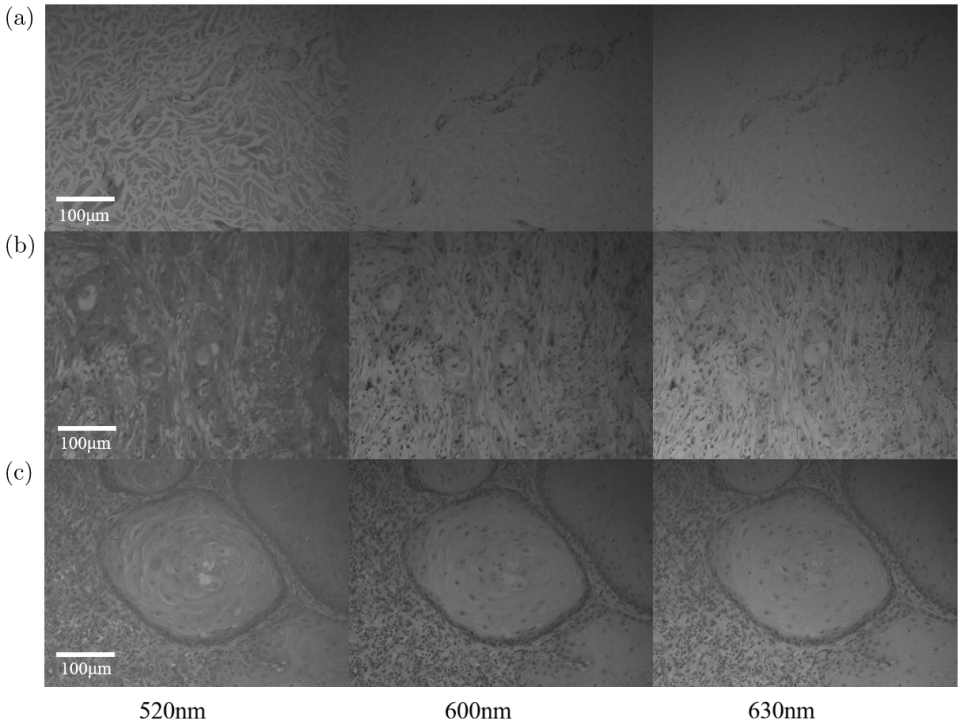


Fig. 5. Multispectral images of 520 nm, 600 nm, and 630 nm with different tissues ((a) Normal tissue (b) Poorly-differentiated cSCC, and (c) Highly-differentiated cSCC).

specimens. This acquisition methodology generated a highly diversified dataset that significantly improved the performance of our deep learning-based classification model, enabling precise discrimination between normal tissue and various cSCC differentiation grades. The established multispectral approach provides a robust framework for improving diagnostic accuracy in histopathological analysis through spectral feature engineering.

3.2. *Comparison of classification performance of each model*

The experiments in this study were conducted on the PyCharm platform. To ensure reproducibility, deep learning models were developed and trained in a consistent software environment using Python 3.8.3, PyTorch 1.10.0, and Torchvision 0.11.1. GPU acceleration was supported by CUDA 11.3 and cuDNN 8.2.0. The computational setup included an NVIDIA GeForce RTX 2070 GPU (8 GB VRAM), Intel Core i7-9700K CPU, and 32 GB of RAM, which collectively provided the necessary hardware acceleration for model training and evaluation.

We trained and tested the collected multispectral data on five models: DenseNet,[37] EfficientNet,[38] Swin-Transformer,[39] ResNet,[40] and MobileViT. To ensure a fair comparison, a consistent training strategy was applied across all architectures. All models were trained for 100 epochs using the Adam optimizer. The learning rate was kept constant throughout the training process, without the use of a learning rate scheduler. The learning rate and batch size were carefully tuned for each model: a batch size of 16 was used for most models, while the Swin-Transformer employed a batch size of 32 due to its architectural characteristics. The detailed hyperparameter configurations are summarized in Table 4. The baseline models were trained to minimize the cross-entropy loss, whereas our optimized

Table 4. Parameter settings for each model.

	Batch size	*lr*
ResNet	16	0.0001
EfficientNet	16	0.01
DenseNet	16	0.001
Swin-Transformer	32	0.0002
MobileViT	16	0.0002

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MobileViT utilized the proposed focal loss. For all experiments, the model weights from the epoch with the highest validation accuracy were selected for final evaluation on the independent test set.

As listed in Table 5, we conducted a comprehensive evaluation of the five mainstream models under identical experimental conditions. The results demonstrate that MobileViT achieved superior overall performance with an accuracy of 91.70%, outperforming other architectures, including EfficientNet (90.70%), ResNet (89.80%), Swin-Transformer (88.00%), and DenseNet (81.60%).

To gain deeper insight into the models' discriminatory capabilities at the category level, we performed a per-class analysis using the AUC metric, as presented in Fig. 6. The analysis revealed that all models achieved perfect separation (AUC = 1.00). For the more challenging highly differentiated and poorly differentiated categories, all architectures maintained strong performance with AUC values consistently above 0.98, although with minimal inter-model variation.

Table 5. Accuracy test results of each model.

Model	Accuracy (%)
DensNet	81.60
EfficientNet	90.70
Swin-Transformer	88.00
ResNet	89.80
MobileViT	91.70

Despite comparable AUC performance across models, we selected MobileViT as our baseline for further optimization based on three principal considerations. First, its highest overall accuracy indicates the most effective feature representation learning across all categories. Second, its hybrid architecture combining CNN's local feature extraction with Transformer's global modeling capabilities offers greater flexibility for subsequent architectural enhancements. Third, MobileViT provides favorable computational efficiency compared to pure Transformer models, making it more suitable for practical deployment scenarios.

Notably, MobileViT's AUC values for both the highly differentiated and poorly differentiated categories were only 0.98, suggesting a potential class imbalance within the dataset where the normal category may dominate the accuracy metric. Therefore, our optimization efforts will focus not only on improving the overall accuracy, but also,

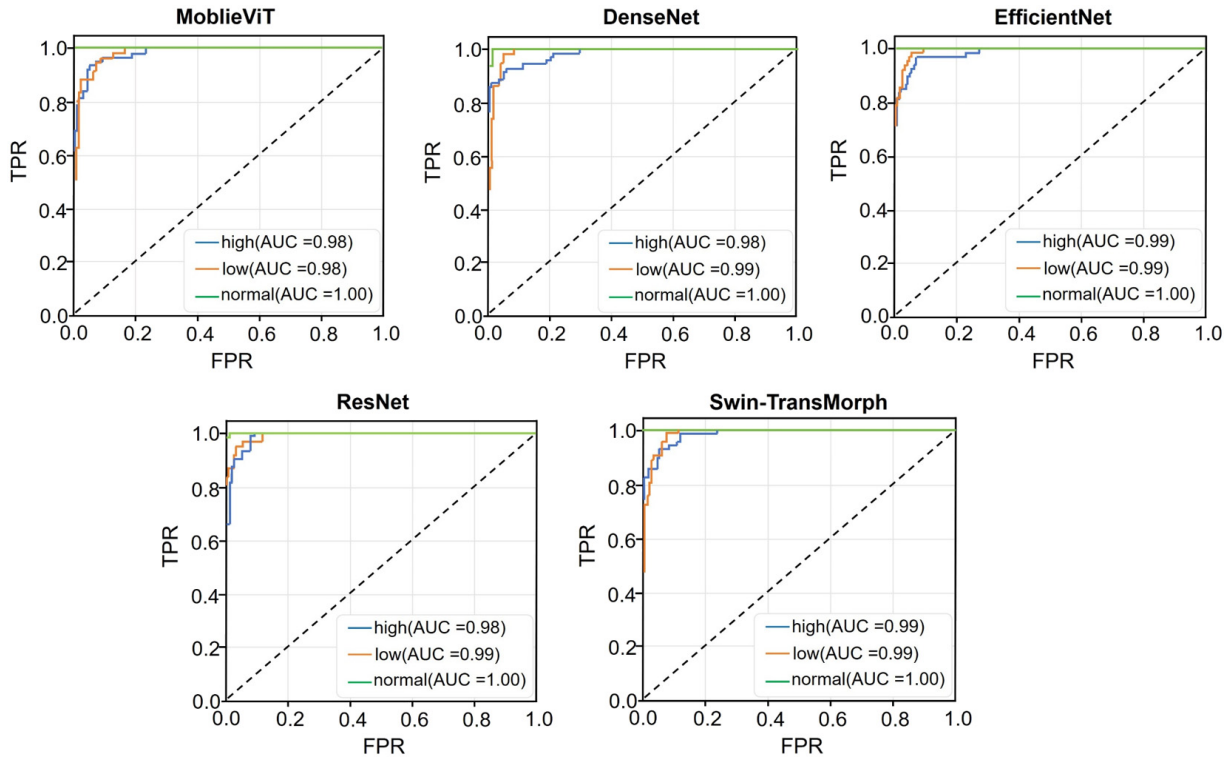


Fig. 6. ROC curves and AUC values for different models.

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more importantly, on enhancing the model's ability to discriminate between the highly differentiated and poorly differentiated categories. Addressing this performance bottleneck aims to develop a more comprehensive and balanced classification system.

3.3. *Optimization and improvement of the model*

3.3.1. *The negative detection performance of the optimized MobileViT model*

In this set of experiments, we employed the optimization strategies introduced in Chapter 2 to evaluate their impact. We input the data-augmented multispectral images into our proposed optimized model and the original MobileViT model to compare their performance. The optimized model is referred to as optimized MobileViT. The class weight values were calculated based on the class proportions of the training set: highly differentiated cSCC was assigned a weight of approximately 0.34, poorly differentiated cSCC of approximately 0.38, and normal tissue of approximately 0.28, with the focal factor set to 2.

After training the models before and after optimization for 100 epochs, the training loss functions are shown in Fig. 7. The x-axis represents the number of epochs, and the y-axis represents the value of the loss function. The results indicate that the loss values of the models both decreased with the increase in epochs. In the initial stage, there was not much difference in performance between the two models, and the final loss rates converged within 0.05, indicating that both models have good stability and efficiency in classifying multispectral microscopic pathological images. The original MobileViT model (blue curve) entered a stable state after 40 epochs, whereas the optimized MobileViT (red curve) did so between 20 and 30 epochs, indicating that our model converges faster and has better classification performance.

By evaluating the classification accuracy, precision, recall, $F1$-score, and specificity of the models before and after optimization, we obtained the data presented in Table 6. The data in the table show that the original MobileViT model performed excellently in the task of negative prediction, achieving an accuracy of 91.7% in distinguishing between cancerous and normal tissues. It demonstrated a high level of identification capability for normal tissues, with precision, $F1$-score, recall, and specificity all above 98%, almost completely avoiding false negatives, and thereby significantly improving negative exclusion. However, its performance was insufficient for classifying highly differentiated and poorly differentiated cSCC. After optimization, the negative prediction performance of the model reached 100% in all aspects, indicating that the accurate classification of normal tissue samples was consistently maintained, and the optimized model also improved the identification of highly differentiated and poorly differentiated tissues.

For normal tissues, the optimized model achieved 100% accuracy, $F1$-score, and specificity, confirming perfect performance in the negative detection task. For highly-differentiated cSCC, the model's precision and $F1$-score showed significant improvement, with increases of 3.20% and 0.50%, respectively, indicating an enhanced classification capability for highly-differentiated cSCC samples and a significant



Fig. 7. The loss function curve graph before and after optimization.

Table 6. Comparison of each index before and after optimization.

Evaluation dimension	Category	MobileViT (%)	Optimized MobileViT (%)	Amplitude (%)
Recall	High	79.70	78.30	−1.40
	Low	93.30	96.70	3.40
	Normal	100.00	100.0	0.00
Precision	High	93.20	96.40	3.20
	Low	81.20	79.50	−1.70
	Normal	98.90	100.0	1.10
$F1$-score	High	85.90	86.4	0.50
	Low	86.80	87.3	0.50
	Normal	99.40	100.0	0.60
Specificity	Normal	98.90	100.0	1.10
Accuracy		91.70	92.80	1.10

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reduction in misdiagnoses. For poorly-differentiated cSCC, the recall rate increased from 93.3% to 96.7%, ensuring that most poorly-differentiated cSCC samples were not missed, and the $F1$-score improved from 86.8% to 87.3%. Although the precision decreased by 1.70%, this minor trade-off is acceptable given the substantial gain in recall, which ensures that more true poorly differentiated cSCC samples are identified.

Figure 8 displays a stacked confusion matrix of the models before and after optimization. The results show that before optimization, out of 69 samples with true labels as highly differentiated cSCC, 55 were correctly predicted as highly differentiated cSCC, while 13 were misclassified as poorly differentiated cSCC and 1 was misclassified as normal tissue. After optimization, the probability of correct prediction increased, and crucially, no tumors were misclassified as normal tissue. This indicates that, while perfect identification of normal tissue was maintained, the risk of missing cancer patients was eliminated. Additionally, in the classification of poorly differentiated cSCC, before optimization, out of 60 samples with true labels as poorly differentiated cSCC, 56 were correctly predicted, and 4 were misclassified as highly differentiated cSCC. After optimization, the number of correct predictions increased to 58, and the number of misclassifications as highly differentiated cSCC decreased to 2, indicating that the model's ability to recognize poorly differentiated cSCC was optimized, reducing confusion with highly differentiated cSCC.

These results demonstrate that the optimized MobileViT model performed excellently across most evaluation dimensions. It achieved significant improvements, particularly in recall for the low-differentiation category and precision for the high-differentiation category, alongside an overall accuracy increase. This proves the effectiveness of our optimization strategies. Although there were minor performance variations in some categories, overall, the optimized model showed superior comprehensive performance in handling complex tasks, providing strong support for further practical applications.

3.3.2. *The optimized model was compared with the baseline model in negative tests*

We trained and tested the collected multispectral data using the optimized model and a series of baseline models. The corresponding results are shown in Fig. 9. The figure shows the overall accuracy of these models, and it is evident that the overall accuracy of our model, at 98%, is significantly higher than that of the other models.

Table 7 presents the precision, recall, $F1$-scores, and specificity data for each model. Performance analysis indicates that optimized MobileViT exhibits a comprehensively leading performance in histopathological classification tasks. The model achieved a breakthrough in the identification of normal tissues, with precision, recall, $F1$-score, and specificity all reaching 100%, significantly outperforming comparative models. For instance, while ResNet achieved

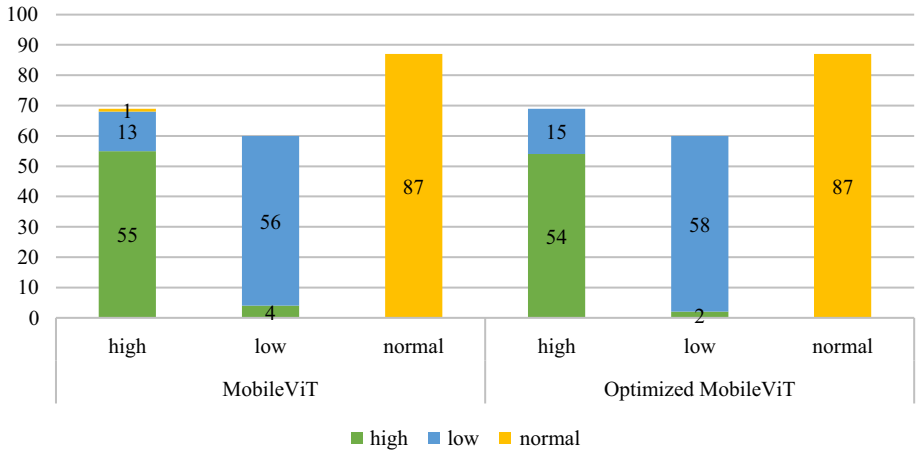


Fig. 8. Stacked diagrams of confusion matrices for each model (Green represents the "high" category. Blue represents the "low" category. Orange represents the "normal" category. Each model's bar is divided into these three colored sections, with the height of each section indicating the quantity or percentage of that category for the model. By observing the proportions of these colors, one can compare the performance of different models across the various categories.).

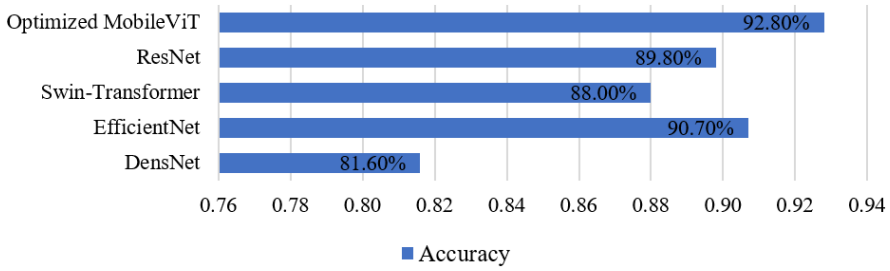


Fig. 9. Bar charts of the accuracy rates of each model.

Table 7. Model metric comparisons for the five models.

Optimized MobileViT	Precision (%)	Recall (%)	*F*1-score (%)	Specificity (%)
High	96.4	78.3	86.4	98.6
Low	79.5	96.7	87.3	90.4
Normal	100.0	100.0	100.0	100.0
DenseNet	Precision	Recall	F1 score	Specificity
High	85.5	68.1	75.8	94.6
Low	72.2	86.7	78.8	87.2
Normal	97.8	100.0	98.9	98.4
EfficientNet	Precision	Recall	F1 score	Specificity
High	96.2	73.9	83.6	98.6
Low	78.4	96.7	86.6	89.7
Normal	97.8	100.0	98.9	98.4
ResNet	Precision	Recall	F1 score	Specificity
High	98.0	69.6	81.4	99.3
Low	75.0	100.0	85.7	87.2
Normal	98.9	98.9	98.9	99.2
Swin-Transformer	Precision	Recall	F1 score	Specificity
High	92.2	68.1	78.3	97.3
Low	76.7	93.3	84.2	89.1
Normal	94.6	100.0	97.2	96.1

the highest *F*1-score (98.9%) among the alternatives, optimized MobileViT's perfect scores demonstrate its ability to completely eliminate the risk of misdiagnosing normal tissue. For highly-differentiated tissues, its precision (96.4%) and specificity (98.6%) were second only to those of EfficientNet and ResNet. In identifying poorly-differentiated tissues, it achieved the leading recall (96.7%, tied with ResNet) and the highest *F*1-score (87.3%, exceeding EfficientNet by 0.7%). The comprehensive *F*1 average across the three tissue types reached 91.2% (an improvement of 2.8%/1.5% over ResNet/EfficientNet, respectively), validating its robustness and generalization capabilities as the optimal model.

Figure 10 presents the confusion matrices for all the compared models. By scrutinizing the true-positive, false-positive, true-negative, and false-negative entries, we quantitatively assessed each model's performance on multispectral histopathological image classification. The confusion-matrix analysis demonstrated that optimized MobileViT achieved the highest diagnostic safety and subtype-discrimination accuracy. In the negative-sample exclusion task, optimized MobileViT correctly identified all 87 normal cases without misclassifying any tumor as normal, yielding zero false negatives and zero false positives. In contrast, ResNet misclassified one cancer case as normal, whereas DenseNet and EfficientNet each made two such errors. Thus, optimized MobileViT significantly outperformed the other four models in excluding malignancy. Regarding cancer-subtype differentiation: (1) Among 69 highly differentiated cases, optimized MobileViT misclassified 15 as poorly differentiated, whereas ResNet and DenseNet misassigned 20 each, and EfficientNet and Swin-TransMorph misassigned 16 and 17, respectively. (2) Among 60 poorly differentiated cases, optimized MobileViT misclassified only 2 as highly differentiated, whereas DenseNet and Swin-Transformer misclassified 8 and 4, respectively. These results indicate that optimized MobileViT also excels at recognizing poorly differentiated tissue.

Collectively, optimized MobileViT not only guarantees reliable exclusion of malignancy but also refines the cSCC subtype classification, reducing confusion between highly differentiated and poorly differentiated lesions. Consequently, it provides pathologists with a more precise grading reference while avoiding the critical error of labeling cancerous tissue as benign.

4. Discussion

This study proposes a novel technical approach, namely, AI-Enhanced Multispectral Technology, which achieves 100% negative case exclusion for

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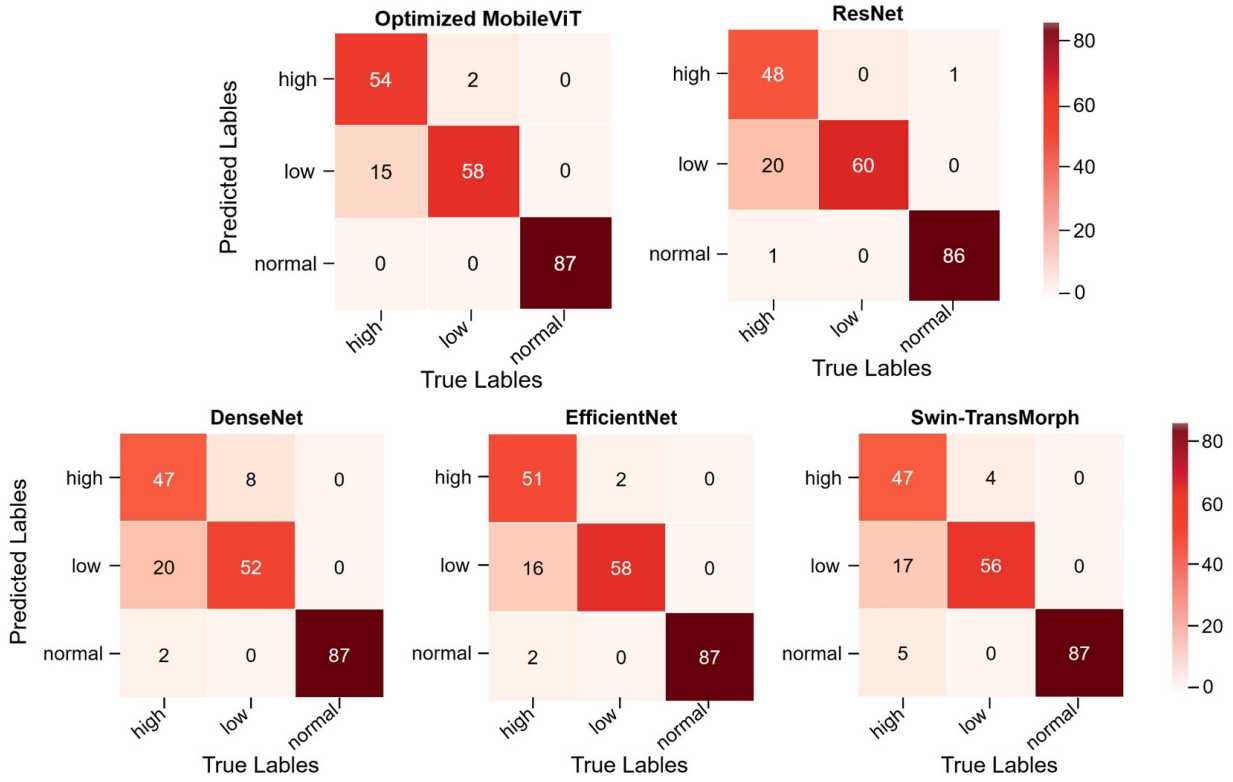


Fig. 10. Confusion matrices of the five models.

pathological diagnosis. Compared to conventional RGB imaging, MSI has demonstrated significant advantages in feature extraction. To leverage these advantages, we selected three key spectral bands — 520 nm, 600 nm, and 630 nm — based on the optical characteristics of H&E staining, which were optimized for imaging specific tissue structures: the 520 nm band effectively enhanced contrast in eosin-stained regions, the 600 nm band highlighted extracellular matrix features, and the 630 nm band provided optimal contrast for nuclear details. This multiband synergistic strategy overcomes the signal aliasing limitations inherent in RGB imaging, providing richer and more accurate feature information for subsequent deep learning analysis.

The MSI approach demonstrated in this study offers distinct practical advantages over hyperspectral imaging (HSI) for clinical pathological screening. The primary benefits lie in its higher imaging throughput, reduced system complexity, and significantly reduced data redundancy. In contrast, HSI provides more refined spectral information, which often requires lengthy acquisition times (ranging from minutes to tens of minutes per FOV) because of the need to capture hundreds of spectral bands, consequently generating massive datasets that pose significant challenges for storage and computational processing.

With regard to the feature extraction capabilities, this study evaluated the performance of multiple deep learning architectures. A critical synthesis highlights their distinct advantages and drawbacks, which underpinned our choice of the optimized MobileViT. Conventional CNNs (ResNet, DenseNet, EfficientNet) offer strong local feature extraction and parameter efficiency — with DenseNet enabling feature reuse, EfficientNet balancing model scaling, and ResNet mitigating gradient vanishing — but they are inherently limited in modeling global contextual relationships within tissue architecture. In contrast, pure Transformers (Swin-Transformer) excel at capturing long-range dependencies, yet they demand high computational resources and are prone to overfitting on smaller datasets. Notably, our optimized MobileViT model strikes a deliberate balance by combining the local feature extraction capability of CNNs with the global modeling

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strength of Vision Transformers (ViTs). This hybrid architecture, combined with class-imbalance-aware focal loss, directly addresses the core challenge of squamous cell carcinoma (cSCC) grading: the need to resolve fine-grained cytological details and broad tissue patterns simultaneously. This is the key reason for its superior performance in classifying cSCC cases with varying differentiation grades.

5. Conclusion

This study innovatively developed a pathology-negative screening algorithm based on MSI, which achieved 100% accuracy in excluding negative cases during tissue section analysis. Utilizing three characteristic wavelengths — 520 nm, 600 nm, and 630 nm — the system precisely captures wavelength-specific absorbance differences associated with malignant states, enabling quantitative discrimination between normal and pathological tissues. By leveraging multispectral data analysis, the proposed method significantly enhances the efficiency and reliability of negative sample identification, offering an effective and reliable decision support tool for pathological diagnosis. Integrated with an optimized MobileViT deep learning architecture, our methodology achieved 100% specificity in negative sample identification ("negative exclusion rate"), effectively reducing false positives and diagnostic omissions while providing high-confidence decision support for clinical "negative exclusion" protocols. The superiority of this framework stems not only from MSI's enhanced biomarker discovery capabilities, but also from improved model interpretability through AI-driven multimodal analysis (spectral-spatial feature fusion), which synergistically supports high-precision diagnostic outcomes. Furthermore, this paradigm offers a transferable potential for malignancies with distinct spectral signatures. Notable examples demonstrate the clinical validity of MSI-AI integration, with the AstroPath MSI platform achieving 91% accuracy in predicting immunotherapy response for melanoma when combined with automated intelligent algorithms.[28] Similarly, narrowband MSI incorporating dynamic threshold segmentation improved cervical precancerous lesion detection rates to 93.5%.[41] These successful implementations provide compelling evidence for the generalizability of combined MSI and AI technologies across diverse oncological applications. This technology holds transferable potential for the early screening of epidermal tumors such as basal cell carcinoma and oral cSCC, as well as for molecular typing and efficacy prediction of solid tumors such as breast cancer and colorectal cancer. In particular, they have broad clinical application prospects for rapid diagnosis scenarios with high timeliness requirements.

However, as with all technologies, continuous improvement and adaptation to emerging challenges in clinical diagnostics are indispensable. To further enhance the role of AI in pathology and medical imaging, it is necessary to address several advancements and challenges.

One important area of future development lies in improving the generalization of AI models to different clinical environments and datasets.[42] Although MobileViT performs excellently in the current dataset, there may be challenges when applied to datasets with different characteristics, such as those from different hospitals or populations. This issue of domain adaptation is crucial for ensuring the robustness and applicability of AI in various healthcare settings. Incorporating larger, more diverse datasets, including multi-center and multi-population data, will help AI models become more adaptable and accurate across different clinical contexts. Additionally, ongoing updates and retraining with real-world data will allow AI models to stay relevant as medical practices evolve.[43]

Another avenue for improvement is the integration of multimodal data. While this study focuses on MSI data, combining AI-driven imaging analysis with clinical data, patient history, and genomic data has the potential to provide a more comprehensive and holistic approach to diagnosis. For example, integrating genomic markers, patient-specific data, and immunohistochemistry findings can further refine the accuracy of the AI model, enabling a more personalized treatment approach. The ability to incorporate such multi-dimensional data may offer a more nuanced understanding of the tumor's behavior and characteristics, leading to more accurate predictions for poorly differentiated cSCC and highly differentiated cSCC. Future models could also incorporate real-time patient feedback and longitudinal data, which would enable dynamic and continuous improvements in clinical decision-making.

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In terms of model efficiency, the computational demands of AI models should also be considered. As models become more sophisticated and require larger datasets for training, ensuring that they are computationally efficient and can be deployed on available medical devices is critical. Advances in hardware, such as edge computing and cloud-based systems, can provide solutions for deploying these models efficiently, even in resource-limited settings.

In summary, while MobileViT shows excellent performance in negative detection for normal tissue and cSCC subtypes, there is ample room for further advancement. As AI continues to evolve, it will likely become an indispensable tool in pathology, assisting clinicians in making more accurate, efficient, and personalized diagnoses. Ultimately, the goal is to develop AI models that not only support but also enhance the expertise of pathologists, leading to better patient outcomes and more efficient healthcare systems.

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Conflict of Interest

The authors declare that there are no conflicts of interest relevant to this paper.

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